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MILITARY SPECIFICATION

BOMB, LOADING PRACTICE BDU-32/E

1. SCOPE. This specification covers the requirements for one type of loading practice bomb designated as the BDU-32/E, hereafter referred to as the practice bomb.

2. APPLICABLE DOCUMENTS. The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-B-26195	Boxes, Wood-Cleated, Skidded, Load-Bearing Base
MIL-B-27000	Bombs, Practice, General Specification For
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods

DRAWINGS

Air Force

61D14133	Cover Assembly, Radome
63C13046	Parachute Assembly
64J13100	Bomb, Practice Loading BDU-32/E
64D13104	Ring Assembly, Forward Clamp

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64D13111	Clamp, Aft
64D13122	Diagram, Schematic, Electrical
64C13123	Receptacle Assembly, Center Case
64J13127	Aft Section, Assembly
64E14401	Nose Assembly
64J14404	Case Section Assembly

(Copies of documents required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 GENERAL SPECIFICATION. The requirements of MIL-B-27000 shall apply as requirements of this specification with the exceptions and additions specified herein. In case of conflict between the two specifications, this specification shall govern.

3.2 PREPRODUCTION. This specification makes provisions for preproduction testing.

3.3 COMPONENTS. The practice bomb shall consist of the following major components:

	ITEM	PART NUMBER	NOMENCLATURE	QUANTITY
	a.	61D13127	Cover Assembly, Radome	1
	b.	63C13046-4	Parachute Assembly (Dummy)	1
	c.	64D13104	Ring Assembly, Forward Clamp	1
	d.	64D13111	Clamp, Aft	1
	e.	64J13127	Aft Section, Assembly	1
	f.	64E14401	Nose Assembly	1
	g.	64J14404	Case Section Assembly	1
*	h.	64D13131	Cable Assembly, Special Purpose, Electrical HRU-561/E	1

*This item may be procured separately.

3.4 DESIGN AND CONSTRUCTION. The practice bomb and components shall be designed and constructed in conformance to drawing 64J14400 and subsidiary drawings.

3.4.1 INTERCHANGEABILITY. Parts functionally and dimensionally interchangeable shall be identified by the same part number. When parts are not interchangeable, the criteria listed in MIL-D-70327 shall apply and new drawings shall be prepared as necessary.

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3.4.2 WEIGHT. The weight of the practice bomb shall be 500 pounds ± 15 pounds.

3.4.3 CENTER OF GRAVITY. Center of gravity of the practice bomb shall be station 54.60 ± 1 inch.

3.4.4 LOCATION REFERENCE. Station 0 shall be the forward surface of the nose of the practice bomb.

3.4.5 PITCH AND YAW MOMENT OF INERTIA. Pitch and yaw moment of inertia of the practice bomb shall be 430000 lb-in^2, plus 10 percent minus zero.

3.5 FINISH. Finish requirements for the practice bomb shall be as specified in MIL-B-27000.

3.5.1 FINAL FINISH COAT. Final finish coat shall be as specified in the engineering drawings.

3.6 PERFORMANCE. Performance of the practice bomb shall consist of:

a. Simulation of the electrical and physical characteristics of the parent weapon.

b. Capability of withstanding the loads and stresses incident to transportation, aircraft loading and flight.

3.7 ENVIRONMENTAL REQUIREMENTS. The practice bomb shall be capable of meeting the performance requirements of this specification during or after, as applicable, exposure to the following environmental conditions:

a. High Temperature to 71°C (160°F).

b. Low Temperature to -54°C (-65°F).

c. Temperature shock incident to service use.

d. Humidity incident to storage, transportation and handling in a damp climate.

e. Salt spray likely to be encountered in storage or transportation near a large body of salt water.

f. Shock likely to be encountered during transportation or handling in service use.

g. Vibration incident to transportation and flight in carrier aircraft.

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3.8 IDENTIFICATION OF PRODUCT. The practice bomb and all parts and assemblies shall be marked in accordance with MIL-STD-130.

3.9 GOVERNMENT FURNISHED PROPERTY. Unless otherwise specified in the contract, the following item shall be furnished to the contractor, upon his request, as Government furnished property for the purpose of fabrication of the practice bomb:

ITEM NUMBER	PART NUMBER	ITEM DESCRIPTION	QUANTITY
1	63C13046-4	Parachute Assembly (Dummy)	1 per practice bomb

3.10 WORKMANSHIP. Workmanship shall be as specified in MIL-B-27000 (USAF).

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS. Inspection and testing of the practice bomb shall be classified as follows:

a. Preproduction Tests

b. Acceptance Tests

4.2 PREPRODUCTION TESTING

4.2.1 PREPRODUCTION TEST SAMPLES. The preproduction test units shall be representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's facility under the supervision of the procuring activity.

4.2.1.1 PREPRODUCTION TEST REPORTS. In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report in accordance with MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

4.2.2 PREPRODUCTION TESTS. Preproduction tests shall consist of all tests described in "4.5 TEST METHODS."

4.3 ACCEPTANCE TESTS. Acceptance tests shall consist of the following individual tests:

a. Examination of Product

b. Conformance to Drawings

c. Functional Tests

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4.4 TEST CONDITIONS. Test conditions shall be as required for each specific test.

4.5 TEST METHODS

4.5.1 EXAMINATION OF PRODUCT. Each practice bomb shall be examined for strict conformance to the requirements of this specification for which tests are not specified. Failure to meet these requirements shall be cause for rejection.

4.5.2 CONFORMANCE TO DRAWINGS. Each practice bomb shall be inspected for conformance to the drawings and specifications and standards specified therein. Failure to conform to these criteria shall be cause for rejection.

4.5.3 FUNCTIONAL TESTS

4.5.3.1 CONTINUITY. Connect the electrical components as shown in drawing 64D13122. Using a 28 volt d.c. external power source and a test lamp, test for continuity between pins A and J of 64C13123 receptacle. Indication of "No Continuity" shall be cause for rejection.

4.5.3.2 TIMERS. The TA and TB timers shall be checked for the following:

a. Either timer may be set individually to settings of 0 to 59.

b. In no case shall it be possible to set the TA timer at a higher reading than the TB timer when both timers are used.

c. When the TB timer is set at any setting from 0 to 59 and the TA timer is moved to a higher setting, the TB timer shall move to the TA setting. Failure to meet these requirements shall be cause for rejection.

4.5.3.3 SELECTOR SWITCHES. Each selector switch shall be operated over its full range of positions. The switches shall move freely in both directions and remain positively in each setting. It shall be impossible to move the lower selector switch to the third (extreme right) position without releasing the stop screw. Failure to satisfy these requirements shall be cause for rejection.

4.5.4 STATIC LOAD TEST. Static load testing of the practice bomb shall consist of procedure I of the suspension test specified in MIL-B-27000(USAF). This test shall be performed once simulating 30 inch suspension system provisions and once simulating 14 inch

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suspension system provisions. Rejection provisions of MIL-B-27000 (USAF) shall apply. Design limit loads shall be:

Up	4165 pounds
Down	3675 pounds
Lateral	980 pounds

Longitudinal loading shall not be performed.

4.5.5 LUG TEST

4.5.5.1 TEST SET UP. The representative bomb shall be secured to a test stand by straps of sufficient width to prevent compression damage.

4.5.5.2 TEST PROCEDURE. Tensile loads, applied by any convenient means shall be applied to the lugs in the following manner:

a. Forward 30 inch suspension (folding) lug.

(1) 16134 pounds at an angle of 1.2 degrees right of vertical. Physical damage or permanent deformation shall be cause for rejection.

(2) 7260 pounds at an angle of 66 degrees right of vertical. Physical breaking or tearing out of the lug shall be cause for rejection. Permanent deformation at this load is permissible.

b. Forward 14 inch suspension (threaded) lug.

(1) 20,100 pounds at an angle of 1.5 degrees aft of vertical. Physical damage or permanent deformation shall be cause for rejection.

(2) 26,290 pounds at an angle of 10.2 degrees forward of vertical. Physical failure or tearing out of the lug shall be cause for rejection. Permanent deformation at this load is permissible.

c. Aft lug (30 and 14 inch suspension).

(1) 14850 pounds at an angle of 27.3 degrees aft of vertical. Physical damage or permanent deformation of the lug shall be cause for rejection.

(2) 21,900 pounds at an angle of 9 degrees forward of vertical. Breaking or tearing out of the lug shall be cause for rejection. Permanent deformation at this load is permissible.

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4.5.6 TAIL FIN TEST

4.5.6.1 TEST SET UP. The test set up shall be in accordance with MIL-B-27000.

4.5.6.2 TEST PROCEDURE

4.5.6.2.1 LOAD APPLICATION. Concentrated loads shall be applied at station 103.36, 3.28 inches outboard of the intersection of the bomb skin and tail fin in the manner specified in MIL-B-27000.

4.5.6.2.2 SEQUENCE AND INTENSITY. The loads shall be applied in the following steps:

a. Preload the fins to 720 pounds and establish a zero reference for deflection.

b. Increase the load on each fin to 1080 pounds.

c. Increase the loads on each fin in increments of 110 pounds to 2300 pounds. Measure fin deflection at each load increment up to and including total load.

d. Reduce loads to 720 pounds. Breaking or permanent deformation in excess of 0.10 inch at outboard edge at station 106.00 at any load increment shall be cause for rejection.

e. Increase the loads to 2440 pounds.

f. Increase the loads in increments of 56 pounds to 3000 pounds. Measure fin deflection at each load increment up to and including the total load.

g. Reduce the loads simultaneously to 720 pounds. Failure during or after loading to 3000 pounds shall be cause for rejection. Permanent deformation is permissible.

4.5.7 WEIGHT TEST. The practice bomb shall be weighed using a commercial crane scale or equivalent. The weight shall be 500 pounds $\pm$ 15 pounds. Weight not falling within these tolerances shall be cause for rejection.

4.5.8 CENTER OF GRAVITY. The center of gravity of the practice bomb shall be determined by any convenient means. The center of gravity shall be at station 54.60 $\pm$ 1 inch. A determined center of gravity not within these tolerances shall be cause for rejection.

4.5.9 MOMENT OF INERTIA TEST

4.5.9.1 FIXTURE AND PREPARATION. The fixture for this test shall be a simple torsion pendulum or bar with an adjustable hanger which can

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be attached to the lugs of the practice bomb. This fixture shall be calibrated by swinging a bar of known weight and calculated moment of inertia. The periods (in seconds/oscillation) obtained from this swinging shall by substitution in the following formula, be used to determine the spring constant of the torsion bar:

$$I_1 = C_Z \frac{P}{\pi} = I_2$$

Where I_1 = measured inertia (slug ft^2)

C_Z = spring constant of torsion bar (ft-lb/radian)

P = period (seconds/oscillation)

I_2 = moment of inertia of hanger

4.5.9.2 TEST PROCEDURE. The practice bomb shall be attached to the hanger and swung, and the formula in 4.5.9.1 shall be used to determine moment of inertia of the bomb. Pitch and yaw moment of inertia shall be 430000 $lb\text{-}in^2$, plus 10 percent, minus 0. Determined moment of inertia not within these tolerances shall be cause for rejection.

4.5.10 ENVIRONMENTAL TESTS. The practice bomb shall be subjected to the following environmental tests in the sequence in which listed:

	TEST	TEST STANDARD	METHOD	PROCEDURES
a.	High Temperature	MIL-STD-810	501	I
b.	Low Temperature	MIL-STD-810	502	I
c.	Temperature Shock	MIL-STD-810	503	I
d.	Humidity	MIL-STD-810	507	I
e.	Salt Fog	MIL-STD-810	509	I
f.	Shock	MIL-B-27000		
g.	Vibration	MIL-STD-810	514	Table 514-1

Where the test procedure requires operation of equipment, the function tests of this specification shall apply. Failure to satisfy the requirements of the function tests during or after as applicable, subjection to the environmental tests shall be considered as failure to pass the environmental tests and cause for rejection.

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5. PREPARATION FOR DELIVERY

5.1 Levels of preservation and packaging shall be level A, B or C as specified (see 6.3).

5.1.1 LEVEL A AND B. Each bomb shall be preserved and packaged in accordance with MIL-P-116, Method III.

5.1.2 LEVEL C. Each bomb shall be preserved and packaged in accordance with the Manufacturer's Commercial practice.

5.2 PACKING. Packing shall be level A, B or C as specified (see 6.3).

5.2.1 LEVEL A. Each bomb preserved and packaged as specified in 5.1.1 shall be individually packed in an exterior container conforming to the requirements of MIL-B-26195, Class 2, Style A, Type I or II. The sides, top and ends shall be cleated panel plywood construction.

5.2.2 LEVEL B. Each bomb preserved and packaged as specified in 5.1.1 shall be individually packed in an exterior container conforming to the requirements of MIL-B-26195, Class 1, Style A, Type I or II. The sides, top and ends shall be cleated panel plywood construction.

5.2.3 LEVEL C. Each bomb shall be packed in exterior shipping container in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification rules and regulations of the common carriers as applicable to the mode of transportation.

5.3 PHYSICAL PROTECTION. Cushioning, anchoring, bracing, and blocking with appropriate test methods shall be in accordance with MIL-STD-1186.

5.4 MARKING. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129. The words "Reuseable Container - Do Not Destroy" shall be stencilled on all sides of exterior containers. Letters shall be one inch high or higher. The nomenclature shall be:

BOMB, LOADING PRACTICE BDU-32/E

6. NOTES

6.1 INTENDED USE. The practice bomb is intended for use in training ground crews in parachute replacement, handling and loading the parent weapon. The practice bomb is non-expendable and not to be dropped from the carrying aircraft.

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6.2 ASSOCIATED EQUIPMENT. The following item used with, but not part of the practice bomb, should be procured separately to its respective drawings as required for specific weapon/aircraft applications:

DRAWING NUMBER	ITEM DESCRIPTION	QUANTITY PER PRACTICE BOMB
64D13131	Cable Assembly, Special Purpose Electrical HRU-561/E	1

6.3 ORDERING DATA. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Whether preproduction tests will be conducted by the procuring activity.

c. Levels of preservation, packaging and packing.

d. Provisions for making the following items available to the Government parachute packing activity for assembly of the parachute in conformance to Drawing No. 63C13046-4:

DRAWING NO.	ITEM DESCRIPTION	QUANTITY PER PARACHUTE
61D14059	Cap, Parachute, Aft End	1
61D14063	Ring, Suspension Line Front	1
61C14118	Tube, Outside	1
61B14061	Pin, Straight, Headless	4
NK520-10R10	Screw	4

6.3.1 PREPRODUCTION TEST SAMPLES. It is expected that the contract or purchase order will specify that a minimum of two units representative of the production practice bomb will be submitted to the procuring activity for test and evaluation. The invitation for bids, contract or purchase order should specify the point of inspection for these tests.

6.4 PRODUCTION DELIVERY. No production practice bomb should be delivered prior to full approval of the preproduction samples.

6.5 DEFINITIONS

a. PREPRODUCTION TESTS. Preproduction tests are those tests performed prior to production on representative samples of the production equipment to demonstrate that the equipment meets the requirements of this specification.

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b. ACCEPTANCE TESTS. Acceptance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the equipment is equivalent in performance and construction to the approved preproduction samples.

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Custodian:
Air Force - 15

Preparing Activity:
Air Force - 15

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